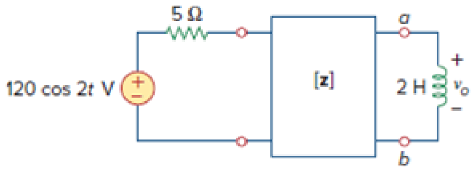
Chapter 19, Problem 16P

Problem

For the circuit in Fig. 19.77, at ω = 2 rad/s, **z**11 = 10 Ω, **z**12 = **z**21 = *j*6 Ω, **z**22 = 4 Ω. Obtain the Thevenin equivalent circuit at terminals *a*-*b* and calculate *v*0.

Figure 19.77



Step-by-step solution

Step 1 of 13

Refer to Figure 19.77 in the text book.

Consider the circuit diagram shown in Figure 1

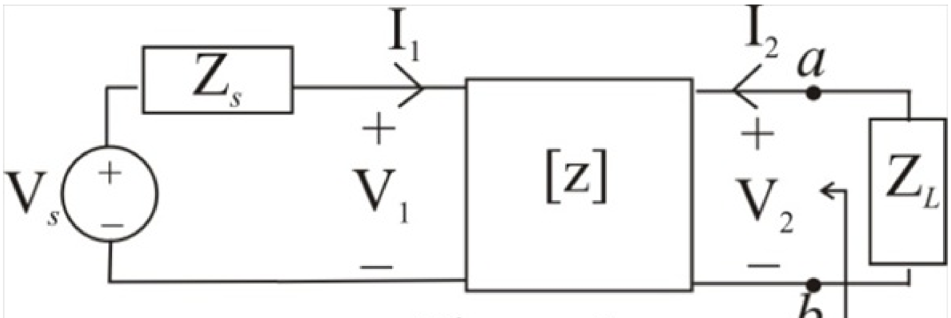


Figure 1

Step 2 of 13

Disconnect the load impedance and redraw the circuit.

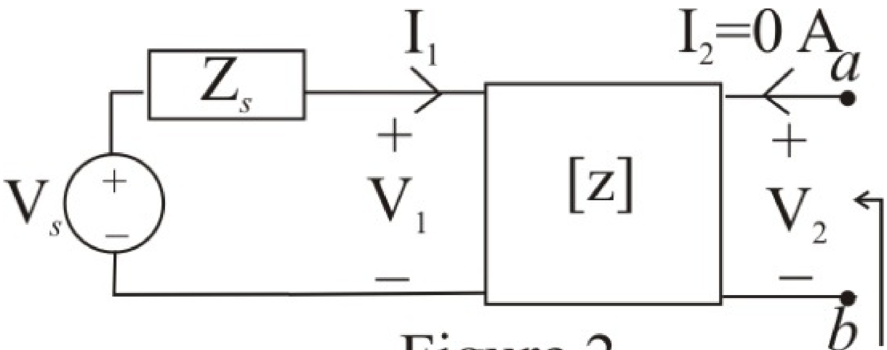


Figure 2

Step 3 of 13

Consider the **z**-parameter equation at output port.

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

Substitute 0 A for $\mathbf{I}_2$.

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}(0)$$
$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1$$
$$\mathbf{I}_1 = \frac{\mathbf{V}_2}{\mathbf{z}_{21}}$$

Therefore, the expression fro the current $\mathbf{I}_1$ is $\frac{\mathbf{V}_2}{\mathbf{z}_{21}}$.

Step 4 of 13

Consider the **z**-parameter equation at input port.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

Substitute $\mathbf{V}_s - \mathbf{I}_1\mathbf{Z}_s$ for $\mathbf{V}_1$ and 0 A for $\mathbf{I}_2$.

$$\mathbf{V}_s - \mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}(0)$$
$$\mathbf{V}_s - \mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1$$
$$\mathbf{V}_s = (\mathbf{z}_{11} + \mathbf{Z}_s)\mathbf{I}_1$$

Substitute $\frac{\mathbf{V}_2}{\mathbf{z}_{21}}$ for $\mathbf{I}_1$.

$$\mathbf{V}_s = (\mathbf{z}_{11} + \mathbf{Z}_s)\left(\frac{\mathbf{V}_2}{\mathbf{z}_{21}}\right)$$
$$\mathbf{V}_2 = \mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\right)$$

Here, the voltage $\mathbf{V}_s$ represents the Thevenin's voltage $\mathbf{V}_{Th}$.

$$\mathbf{V}_{Th} = \mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\right) \quad \text{...... (1)}$$

Therefore, the value of the Thevenin's voltage $\mathbf{V}_{Th}$ across the terminals a-b is,

$$\mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\right).$$

Step 5 of 13

To find the Thevenin's impedance apply a voltage source at the output port and disconnect the source voltage as shown in Figure 3.

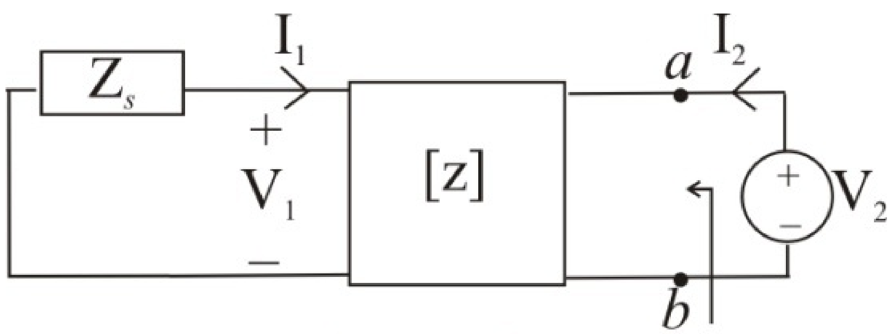


Figure 3

Step 6 of 13

Consider the **z**-parameter equation at input port.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

Substitute $-\mathbf{I}_1\mathbf{Z}_s$ for $\mathbf{V}_1$.

$$-\mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$
$$-\mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$
$$(\mathbf{z}_{11}+\mathbf{Z}_s)\mathbf{I}_1 = -\mathbf{z}_{12}\mathbf{I}_2$$
$$\mathbf{I}_1 = -\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$$

Therefore, the expression for the current $\mathbf{I}_1$ is $-\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$.

Step 7 of 13

Consider the **z**-parameter equation at output port.

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

Substitute $-\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$ for $\mathbf{I}_1$.

$$\mathbf{V}_2 = \mathbf{z}_{21}\left(-\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2\right) + \mathbf{z}_{22}\mathbf{I}_2$$
$$\mathbf{V}_2 = \left(\mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$$
$$\frac{\mathbf{V}_2}{\mathbf{I}_2} = \mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}$$

The ratio $\frac{\mathbf{V}_2}{\mathbf{I}_2}$ represents the Thevenin's impedance $\mathbf{Z}_{Th}$.

$$\mathbf{Z}_{Th} = \mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s} \quad \ldots\ldots (2)$$

Therefore, the expression fro the current $\mathbf{I}_1$ is $\frac{\mathbf{V}_2}{\mathbf{z}_{21}}$.

Recall equation (2).

$$\mathbf{Z}_{Th} = \mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}$$

Substitute $5\ \Omega$ for $\mathbf{Z}_s$, $10\ \Omega$ for $\mathbf{z}_{11}$, $j6\ \Omega$ for $\mathbf{z}_{12}$, $j6\ \Omega$ for $\mathbf{z}_{21}$ and $4\ \Omega$ for $\mathbf{z}_{22}$.

$$\begin{aligned}\mathbf{Z}_{Th} &= 4 - \frac{(j6)(j6)}{10+5} \\ &= 4 + \frac{36}{15} \\ &= 4 + 2.4 \\ &= 6.4\ \Omega\end{aligned}$$

Therefore, the value of the Thevenin's impedance is, $6.4\ \Omega$.

Step 8 of 13

Consider the value of the voltage source.

$$v = 120\cos 2t \text{ V}$$

Convert to phasor form.

$$\mathbf{V} = 120\angle 0° \text{ V}$$

Therefore, the value of the voltage source in phasor form is $120\angle 0° \text{ V}$

Step 9 of 13

Recall equation (1).

$$\mathbf{V}_{Th} = \mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)$$

Substitute $120\angle 0° \text{ V}$ for V_s, $5\ \Omega$ for $\mathbf{Z}_s$, $10\ \Omega$ for $\mathbf{z}_{11}$ and $j6\ \Omega$ for $\mathbf{z}_{21}$.

$$\begin{aligned}\mathbf{V}_{Th} &= (120\angle 0°)\left(\frac{j6}{10+5}\right) \\ &= (120\angle 0°)\left(\frac{6\angle 90°}{15}\right) \\ &= (8\angle 0°)(6\angle 90°) \\ &= 48\angle 90° \text{ V}\end{aligned}$$

Therefore, the value of the Thevenin's voltage $\mathbf{V}_{Th}$ is $48\angle 90° \text{ V}$.

Step 10 of 13

Calculate the reactance due to the load inductance.

$$\mathbf{X}_L = j\omega L$$

Substitute 2 rad/s for ω and 2 H for L.

$$\begin{aligned}\mathbf{X}_L &= j(2)(2) \\ &= j4\ \Omega\end{aligned}$$

Therefore, the value of the reactance due to inductance is $j4\ \Omega$.

Step 11 of 13

Draw the Thevenin's equivalent circuit diagram.

Step 12 of 13

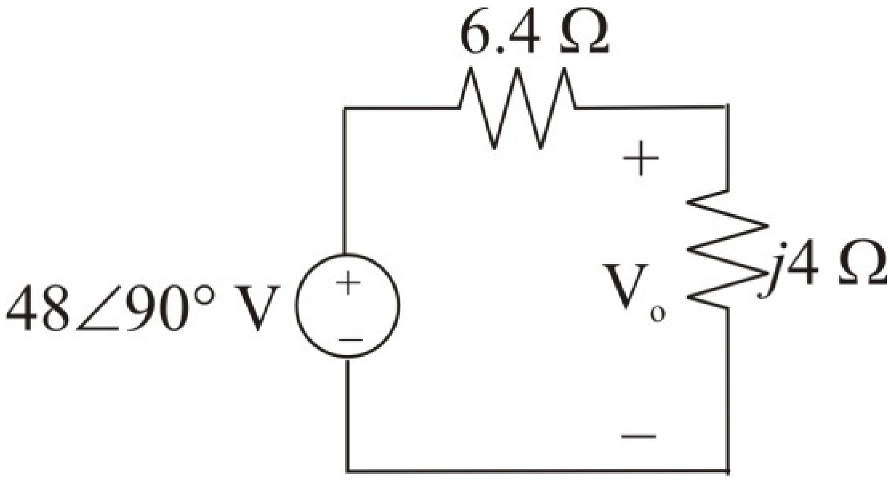


Figure 4

Step 13 of 13

Calculate the voltage $\mathbf{V}_o$ using voltage division rule.

$$\begin{aligned}\mathbf{V}_o &= (48\angle 90^\circ)\left(\frac{j4}{6.4+j4}\right)\\ &= (48\angle 90^\circ)(0.281+j0449)\\ &= (48\angle 90^\circ)(0.53\angle 57.96^\circ)\\ &= 25.44\angle 147.96^\circ \text{ V}\end{aligned}$$

Convert the voltage to time domain.

$$v_o = 25.44\cos(2t+147.96^\circ)\text{ V}$$

Therefore, the value of the output voltage is $\boxed{25.44\cos(2t+147.96^\circ)\text{ V}}$.